

Vibration Analysis for Camera/Monitor Mount and Table Structure

Faculty Review Submission

May 4, 2026

Objective

This document summarizes the vibration analysis performed for the camera/monitor mounting arms and the supporting table structure. Natural frequency calculations were conducted for both arms and the table.

Arm Vibration Analysis

The system is analyzed to ensure the mounted camera and monitor stability. The analysis uses the assumption that the arm may be modeled as a cantilevered beam and simplified as a spring-mass system.

Material and Geometry (Assumed for Arms)

- **Material:** Al 6061-T6 Aluminum
- **Modulus of Elasticity (E):** $E = 69 \text{ GPa}$
- **Yield Strength (σ_y):** $\sigma_y = 240 \text{ MPa}$
- **Moment of Inertia (I):** $I = 1.346 \times 10^{-9} \text{ m}^4$
- **Mass (M):** $M = 0.59\text{kg} + 2.2\text{kg} = 2.79\text{kg}$ (Camera + Mounting Arms)
- **FOS (Factor of Safety):** 2
- **Load (W):** $W = 2 * 2.79 * g \approx 54.7\text{N}$

Arm A Analysis (1.5ft)

Arm A is treated as a simple cantilevered beam with a point load $W = 65 \text{ N}$ at the end.

Deflection and Stiffness Calculation

- **Design Load (W):** $W = 65 \text{ N}$
- **Maximum Deflection ($\delta_{\max}$):**

$$\delta_{\max} = \frac{WL^3}{3EI}$$
$$\delta_{\max} = \frac{54.7 \text{ N} \cdot (0.4572 \text{ m})^3}{3 \cdot (69 \times 10^9 \text{ Pa}) \cdot (1.346 \times 10^{-7} \text{ m}^4)} \approx 0.000188 \text{ m} = 0.188 \text{ mm}$$

- **Equivalent Spring Stiffness (k_A):**

$$k_A = \frac{3EI}{L^3} = \frac{3 \cdot (69 \times 10^9 \text{ Pa}) \cdot (1.346 \times 10^{-7} \text{ m}^4)}{(0.4572 \text{ m})^3} \approx 302,377.55 \text{ N/m}$$

Natural Frequency

- **Total Mass (m):** $m = 2.79$ kg (Camera + Arm System)
- **Natural Frequency ($\omega_{n,A}$):**

$$\omega_{n,A} = \sqrt{\frac{k_A}{m}} = \sqrt{\frac{302,377.55 \text{ N/m}}{2.79 \text{ kg}}} \approx \mathbf{329.21 \text{ Hz}}$$

Arm B Analysis (2ft)

Arm B is analyzed as a cantilevered beam with a point load $W = 65$ N at the end.

Deflection and Stiffness Calculation

- **Maximum Deflection ($\delta_{\max}$):**

$$\delta_{\max} = \frac{WL^3}{3EI}$$
$$\delta_{\max} = \frac{54.7 \text{ N} \cdot (0.6096 \text{ m})^3}{3 \cdot (69 \times 10^9 \text{ Pa}) \cdot (1.346 \times 10^{-7} \text{ m}^4)} \approx 0.00044 \text{ m} = 0.4447 \text{ mm}$$

- **Equivalent Spring Stiffness (k_B):**

$$k_B = \frac{3EI}{L^3} = \frac{3 \cdot (69 \times 10^9 \text{ Pa}) \cdot (1.346 \times 10^{-7} \text{ m}^4)}{(0.6096 \text{ m})^3} \approx 87,738.15 \text{ N/m}$$

Natural Frequency

- **Total Mass (m):** $m = 2.79$ kg
- **Natural Frequency ($\omega_{n,B}$):**

$$\omega_{n,B} = \sqrt{\frac{k_B}{m}} = \sqrt{\frac{87,738.15 \text{ N/m}}{2.79 \text{ kg}}} \approx \mathbf{177.33 \text{ Hz}}$$

Conclusion for Arms

The natural frequencies for both arms are high (~ 329 Hz and ~ 177 Hz accordingly). We are assuming a factory floor vibration environment with dominant frequencies below ~ 100 Hz. Thus, the arms should provide sufficient stability for the mounted camera and monitor, and resonance issues are not expected.

Table Vibration Analysis

The table structure is analyzed for its natural frequencies in the Vertical (Axial) and Lateral (Shear/Bending) directions under different design scenarios.

Table Assumptions and Properties

- **Frame Material:** Al 6061-T6 Aluminum
- **Frame Properties:** $E = 69 \text{ GPa}$, $A = 7.61 \times 10^{-4} \text{ m}^2$, $I = 1.623 \times 10^{-7} \text{ m}^4$, $\rho_{\text{Al}} = 2.06 \text{ kg/m}$
- **Tabletop Material:** HDPE, $\rho \approx 930 \text{ kg/m}^3$, thickness = 6.35 mm
- **Feet:** Rubber pads (Assumed 25 mm diameter, $t = 5 \text{ mm}$), $G_{\text{rubber}} \approx 1.7 \times 10^6 \text{ Pa}$
- **Total Mass (M):** $M_{\text{Al}} \approx 24.04 \text{ kg}$ (frame), $M_{\text{HDPE}} \approx 6.73 \text{ kg}$ (tabletop), $M_{\text{Total}} \approx 35.7 \text{ kg}$ (including components).

Vertical (Axial) Analysis

The table legs are treated as axial springs in parallel.

- **Stiffness of a Single Leg (k_{axial}):**

$$k_{\text{axial}} = \frac{EA}{L} = \frac{(69 \times 10^9 \text{ Pa}) \cdot (7.61 \times 10^{-4} \text{ m}^2)}{0.9 \text{ m}} \approx 5.83 \times 10^7 \text{ N/m}$$

- **Total Stiffness (k_T):** $k_T = 4k_{\text{axial}} = 2.33 \times 10^8 \text{ N/m}$
- **Natural Frequency (f_n):**

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k_T}{M}} = \frac{1}{2\pi} \sqrt{\frac{2.33 \times 10^8 \text{ N/m}}{35.7 \text{ kg}}} \approx \mathbf{407 \text{ Hz}}$$

Lateral (Bending/Shear) Analysis

The lateral analysis considers the stiffness of the legs in bending and the rubber pads in shear. The system is modeled as two springs in series (Leg/Shelf k_S and Rubber k_P).

Initial Design (No Shelves, Small Rubber Pads)

- **Leg Bending Stiffness (k_{leg}):** Assuming fixed-free (cantilever) legs, $k_{\text{leg}} = \frac{3EI}{L^3}$. $k_{\text{leg}} \approx 4.61 \times 10^4 \text{ N/m}$. Total leg stiffness $k_{4\text{legs}} = 1.84 \times 10^5 \text{ N/m}$.
- **Rubber Pad Stiffness (k_{pad}):**

$$k_{\text{pad}} = \frac{GA}{t} = \frac{(1.7 \times 10^6 \text{ Pa}) \cdot (\pi(0.0125 \text{ m})^2)}{0.005 \text{ m}} \approx 1.67 \times 10^5 \text{ N/m}$$

- **Combined Stiffness (k_T per leg):** $k_{T,\text{leg}} = \left(\frac{1}{k_{\text{leg}}} + \frac{1}{k_{\text{pad}}}\right)^{-1} \approx 3.61 \times 10^4 \text{ N/m}$.
- **Natural Frequency (f_n):** $f_n \approx 10.1 \text{ Hz}$. (Too low)

Design with Two Shelves (Small Pads) - Improved Leg Stiffness

The addition of shelves stiffens the legs, increasing k_{leg} . With two equally spaced shelves, the effective leg length L_{eff} is reduced to 0.30 m.

- **Stiffness per Leg (k_S):** $k_S = \frac{12EI}{L_{\text{eff}}^3} = 1.66 \times 10^6 \text{ N/m}$ (Assuming legs are fixed at ends of L_{eff}).
- **Total Stiffness (k_T):** $k_T \approx 6.2 \times 10^5 \text{ N/m}$.
- **Natural Frequency (f_n):** $f_n \approx 20.8 \text{ Hz}$. (Rubber is still limiting)

0.0.1 Design with Two Shelves and Full Rubber Bottom

Changing the rubber pad to cover the whole bottom (area $A = 0.45 \times 1.045 \text{ m}^2$).

- **New Rubber Stiffness (k_P):** $k_P = \frac{GA}{t} \approx 1.65 \times 10^8 \text{ N/m}$.
- **Total Stiffness (k_T):** The combined stiffness is now limited by the stiffened leg/shelf structure.
- **Natural Frequency (f_n):** $f_n \approx 37.2 \text{ Hz}$. (Still considered too low)

Design with Two Shelves, Full Rubber, and HDPE Back Panel

Adding a back panel treats the table as a shear spring, significantly increasing stiffness.

- **Panel Properties (HDPE):** $E = 1.0 \times 10^9 \text{ Pa}$, $\nu = 0.42 \rightarrow G_p \approx 3.52 \times 10^8 \text{ Pa}$. thickness = 6 mm.
- **Back Panel Stiffness (k_{panel}):** Treating as a shear spring, $k_{\text{panel}} = \frac{G_p bt}{h}$, where $h = 0.45 \text{ m}$ for 2 shelves.

$$k_{\text{panel}} \approx 4.29 \times 10^6 \text{ N/m}$$

- **Total Stiffness (k_T):** $k_T = k_{T,\text{frame}} + k_{\text{panel}} = 1.95 \times 10^6 \text{ N/m} + 4.29 \times 10^6 \text{ N/m} = 6.24 \times 10^6 \text{ N/m}$
- **Total Mass (M):** $M = M_{\text{initial}} + M_{\text{panel}} \approx 62.1 \text{ kg} + 4.17 \text{ kg} = 66.3 \text{ kg}$
- **Natural Frequency (f_n):**

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k_T}{M}} = \frac{1}{2\pi} \sqrt{\frac{6.24 \times 10^6 \text{ N/m}}{66.3 \text{ kg}}} \approx 48.8 \text{ Hz}$$

(Better, but not ideal)

Optional Design: Two Shelves, Full Rubber, and Stainless Steel Back Panel

Replacing the back panel with 304 Stainless Steel (SS) for maximum stiffness.

- **Panel Properties (304 SS):** $E = 193 \text{ GPa}$, $G_p \approx 7.42 \times 10^{10} \text{ Pa}$. thickness = 3 mm.
- **SS Back Panel Stiffness (k_{panel}):**

$$k_{\text{panel}} = \frac{G_p bt}{h} \approx 9.06 \times 10^8 \text{ N/m}$$

- **Total Stiffness (k_T):** $k_T = k_{T,\text{frame}} + k_{\text{panel}} = 1.95 \times 10^6 \text{ N/m} + 9.06 \times 10^8 \text{ N/m} \approx 9.08 \times 10^8 \text{ N/m}$ (Dominated by SS panel).
- **Total Mass (M):** $M = M_{\text{initial}} + M_{\text{SS}} \approx 62.1 \text{ kg} + 8.8 \text{ kg} = 70.9 \text{ kg}$
- **Natural Frequency (f_n):**

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k_T}{M}} = \frac{1}{2\pi} \sqrt{\frac{9.08 \times 10^8 \text{ N/m}}{70.9 \text{ kg}}} \approx 181 \text{ Hz}$$

Conclusion for Table

The vertical natural frequency is high (**407** Hz). The lateral natural frequency is the critical design constraint. The final design utilizing ****two shelves, full rubber bottom, and a stainless steel back panel**** achieves a lateral natural frequency of **181** Hz, which is significantly higher and should effectively mitigate vibration issues.